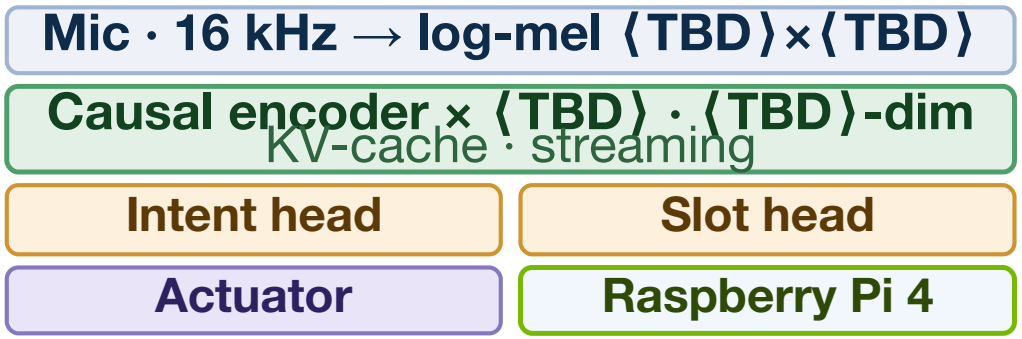
ME2 — Voice Controlled Smart Device

Always-on keyword + intent recognition at the edge · no cloud round-trip

1 October 2026 · ⟨fill: author / team⟩ · Model · Dataset · Training on A100 · Validation on RPi 4

Model



Item	Value
Parameters / weights	⟨TBD⟩ M · ⟨TBD⟩ MB

Validation on the Raspberry Pi 4

Item	Value
Keyword / intent acc	⟨TBD⟩ % / ⟨TBD⟩ %
False-accept rate	⟨TBD⟩ %
Latency p95 / RTF	⟨TBD⟩ ms / ⟨TBD⟩
Runtime	onnxruntime · ⟨TBD⟩ thr

Dataset

Item	Value
Source	⟨fill⟩
Hours / utts	⟨TBD⟩ h / ⟨TBD⟩
Speakers	⟨TBD⟩
Labels	⟨TBD⟩ intents · ⟨TBD⟩ slots

Training on the A100 cluster

Item	Value
Cluster	⟨fill: # A100⟩
Objective	⟨fill: CTC / attn⟩
Optimiser	⟨fill⟩
Steps / loss	⟨TBD⟩ / ⟨TBD⟩

To Be Submitted

GitHub repository	⟨fill: org/repo — public, MIT⟩
Dataset location	⟨fill: Zenodo / HF DOI + access terms⟩
A100 cluster	⟨fill: node ID, # GPUs, wall-clock, seeds⟩
Model weights	⟨fill: release URL + licence⟩

What the reviewer will look for

#	Item	Status
1	Repo public, one-command reproduction	⟨todo⟩
2	Dataset licensed and citable (DOI)	⟨todo⟩
3	Training logs + final checkpoint committed	⟨todo⟩
4	Pi 4 latency reproduced by the posted script	⟨todo⟩
5	Held-out test set, unseen speakers	⟨todo⟩
6	Baseline of comparable size compared	⟨todo⟩